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| Tool Version : Vivado v.2025.1 (win64) Build 6140274 Thu May 22 00:12:29 MDT 2025
| Date         : Sun Dec  7 01:34:49 2025
| Host         : Pavan running 64-bit major release  (build 9200)
| Command      : report_utilization -file top_exact_mult_urbana_utilization_synth.rpt -pb top_exact_mult_urbana_utilization_synth.pb
| Design       : top_exact_mult_urbana
| Device       : xc7s50csga324-1
| Speed File   : -1
| Design State : Synthesized
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Utilization Design Information

Table of Contents

1. Slice Logic

Site Type	Used	Fixed	Prohibited	Available	Util%
Slice LUTs*	82	0	0	32600	0.25
LUT as Logic	82	0	0	32600	0.25
LUT as Memory	0	0	0	9600	0.00
Slice Registers	16	0	0	65200	0.02
Register as Flip Flop	16	0	0	65200	0.02
Register as Latch	0	0	0	65200	0.00
F7 Muxes	0	0	0	16300	0.00
F8 Muxes	0	0	0	8150	0.00

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* Warning! The Final LUT count, after physical optimizations and full implementation, is typically lower. Run opt_design after synthesis, if not already completed, for a more realistic count.
Warning! LUT value is adjusted to account for LUT combining.
Warning! For any ECO changes, please run place_design if there are unplaced instances

1.1 Summary of Registers by Type

Total	Clock Enable	Synchronous	Asynchronous
0	_	-	-
0	_	-	Set
0	_	-	Reset
0	_	Set	-
0	_	Reset	-
0	Yes	-	-
0	Yes	-	Set
0	Yes	-	Reset
0	Yes	Set	-
16	Yes	Reset	-

2. Memory

Site Type	Used	Fixed	Prohibited	Available	Util%
Block RAM Tile	0	0	0	75	0.00
RAMB36/FIFO*	0	0	0	75	0.00
RAMB18	0	0	0	150	0.00

* Note: Each Block RAM Tile only has one FIFO logic available and therefore can accommodate only one FIFO36E1 or one FIFO18E1. However, if a FIFO18E1 occupies a Block RAM Tile, that tile can still accommodate a RAMB18E1

3. DSP

Site Type	Used	Fixed	Prohibited	Available	Util%
DSPs	0	0	0	120	0.00

4. IO and GT Specific

Site Type	Used	Fixed	Prohibited	Available	Util%
Bonded IOB	33	0	0	210	15.71
Bonded IPADs	0	0	0	2	0.00
PHY_CONTROL	0	0	0	5	0.00
PHASER_REF	0	0	0	5	0.00
OUT_FIFO	0	0	0	20	0.00
IN_FIFO	0	0	0	20	0.00
IDELAYCTRL	0	0	0	5	0.00
IBUFDS	0	0	0	202	0.00
PHASER_OUT/PHASER_OUT_PHY	0	0	0	20	0.00
PHASER_IN/PHASER_IN_PHY	0	0	0	20	0.00
IDELAYE2/IDELAYE2_FINEDELAY	0	0	0	250	0.00
ILOGIC	0	0	0	210	0.00
OLOGIC	0	0	0	210	0.00

5. Clocking

Site Type	Used	Fixed	Prohibited	Available	Util%
BUFGCTRL	1	0	0	32	3.13
BUFIO	0	0	0	20	0.00
MMCME2_ADV	0	0	0	5	0.00
PLLE2_ADV	0	0	0	5	0.00
BUFMRCE	0	0	0	10	0.00
BUFHCE	0	0	0	72	0.00
BUFR	0	0	0	20	0.00

6. Specific Feature

Site Type	Used	Fixed	Prohibited	Available	Util%

BSCANE2	0	0	0	4	0.00
CAPTUREE2	0	0	0	1	0.00
DNA_PORT	0	0	0	1	0.00
EFUSE_USR	0	0	0	1	0.00
FRAME_ECCE2	0	0	0	1	0.00
ICAPE2	0	0	0	2	0.00
STARTUPE2	0	0	0	1	0.00
XADC	0	0	0	1	0.00

7. Primitives

Ref Name	Used	Functional Category
LUT6	31	LUT
LUT4	27	LUT
LUT5	24	LUT
IBUF	17	IO
OBUF	16	IO
FDRE	16	Flop & Latch
LUT2	14	LUT
CARRY4	9	CarryLogic
LUT3	6	LUT
BUFG	1	Clock

8. Black Boxes

Ref Name	Used

9. Instantiated Netlists

Ref Name	Used